

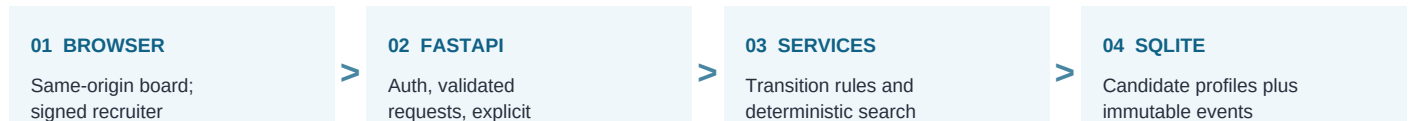
Architecture Summary

One recruiter. One job. Auditable stage changes and read-only search.

REPOSITORY

<https://github.com/diyaavirmani/mini-hiring-pipeline>

Request and data flow



Integrity and concurrency

- Moves run in BEGIN IMMEDIATE transactions and check the expected stage after acquiring the write lock; a stale concurrent request returns a conflict.
- Database constraints and triggers enforce the initial Applied event, one-step forward moves, rejection before Hired, terminal outcomes, immutable history, foreign keys, and a single recruiter.
- Current stage and time in stage are calculated from the newest audit event. The history records actor, transition, reason, and timestamp.

Search and AI boundary

- Known search language is parsed deterministically; supported filters combine, names are fuzzy-ranked, and valid zero matches are distinct from unsupported queries.
- If deterministic parsing cannot interpret a query, optional Gemini may propose structured filters. The app validates them against an allowlist and sends them through the existing read-only search service.
- AI cannot write SQL, edit candidates, or choose stage transitions. Search works without AI credentials for supported queries.

Trade-offs and scope

- SQLite and a same-origin UI keep local setup small for one recruiter. SQLite is not intended here for horizontally scaled instances or sustained multi-user write load.
- AI fallback adds latency, credentials, and provider availability risk. Candidate data in the local SQLite file is not encrypted at rest.
- AWS and Docker are out of scope for this iteration.

Verification

See README.md for fresh-checkout setup, the full test run, the 19-query search evaluation, and browser-flow checks. The suite covers transition and audit integrity, concurrent moves, API access and validation, configuration, AI failure/output validation, and repeat-safe sample seeding.